

Grade 8
Unit 3
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Complete each table. Determine the answer to the division problem in the first row, using whole numbers. Then, convert the whole numbers to scientific notation in the second row.

1.

	Problem	Answer
Whole Numbers	$6,300 \div 100$	63
Scientific Notation	$(6.3 \times 10^3) \div (1 \times 10^2)$	6.3×10^1

2.

	Problem	Answer
Whole Numbers	$600,000 \div 24,000,000$	0.025
Scientific Notation	$(6 \times 10^5) \div (2.4 \times 10^7)$	2.5×10^{-2}

Solve by filling in the blanks for questions 3-5.

3. $\frac{(3.496 \times 10^9)}{(3.8 \times 10^5)} = \frac{3.496}{3.8} \times 10^{9-5} = 9.2 \times 10^3$
 9.2×10^3

4. $\frac{(1.843 \times 10^{-7})}{(4.85 \times 10^{-4})} = \frac{1.843}{4.85} \times 10^{-7-(-4)} = 3.8 \times 10^{-4}$

5. $\frac{(1.892 \times 10^8)}{(4.3 \times 10^3)} = \frac{1.892}{4.3} \times 10^5 = 4.4 \times 10^4$

6. Error Analysis: Marcus and Sara were given the following problem. Their work is shown in the table.

The distance from the sun to the earth is about 150,000,000,000 meters. The speed of sound is about 3.4×10^2 meters per second. About how many seconds does it take for sound from Earth to reach the sun? Write your answer in standard notation, rounding to the nearest tenth.

Marcus's Work	Sara's Work
$\frac{(1.5 \times 10^{11})}{(3.4 \times 10^2)}$ $= \frac{1.5}{3.4} \times \frac{10^{11}}{10^2}$ $= \frac{1.5}{3.4} \times 10^9$ $= 1.9 \times 10^9$	$\frac{(1.5 \times 10^{11})}{(3.4 \times 10^2)}$ $= \frac{1.5}{3.4} \times \frac{10^{11}}{10^2}$ $= \frac{1.5}{3.4} \times 10^{13}$ $= 0.4 \times 10^{13}$

Review Marcus's and Sara's work. Make any corrections to help them learn from their mistakes.

Answers will vary. Both Marcus and Sara incorrectly calculated $\frac{1.5}{3.4}$, which correctly equals approximately 0.44. Sara also added exponents instead of subtracting.

The correct solution is $0.44 \times 10^9 = 4.4 \times 10^8$.

Find the quotient for questions 7-10.

7. $\frac{(4.68 \times 10^{11})}{(2.6 \times 10^3)}$ 1.8×10^8

8. $\frac{(4.644 \times 10^{-8})}{(6.45 \times 10^{-3})}$ 0.72×10^{-5}
 $= 7.2 \times 10^{-6}$

9. $\frac{(6.591 \times 10^7)}{(8.45 \times 10^2)}$ 0.78×10^5
 $= 7.8 \times 10^4$

10. $\frac{(7.7 \times 10^{-8})}{(5.5 \times 10^{-2})}$ 1.4×10^{-6}